Synchrotron Radiation

Particle in magnetic field

$i = 1,2,3$ $\quad \frac{d}{dt}(m\gamma \vec{v}) = \cancel{q\vec{E}} + q\,\vec{v}\times\vec{B}$

$i = 0$ $\quad \frac{d}{dt}(\gamma mc^2) = q\,\vec{v}\cdot\vec{E} = 0$ (energy is conserved)

$\Downarrow$

$\gamma = \text{const} \Rightarrow |\vec{v}| = v = \text{const}$

$$\Rightarrow \gamma m \frac{d\vec{v}}{dt} = q\,\vec{v}\times\vec{B} = q\,\underbrace{\vec{v}_{\parallel}\times\vec{B}}_{\emptyset} + q\,\vec{v}_{\perp}\times\vec{B}$$

let $\vec{v} = \vec{v}_{\parallel} + \vec{v}_{\perp}$, $\vec{v}_{\perp}\cdot\vec{B} = 0$.

$\hookrightarrow$ along $\vec{B}$

$$\Rightarrow \frac{d\vec{v}_{\parallel}}{dt} = 0 , \quad \frac{d\vec{v}_{\perp}}{dt} = \frac{q}{\gamma m}\vec{v}_{\perp}\times\vec{B} = -\frac{q\vec{B}}{\gamma m}\times\vec{v}_{\perp} = -\vec{\omega}_B\times\vec{v}_{\perp}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & 0 \\ 0 & 0 & B \end{vmatrix} = \hat{x}\,v_y B - \hat{y}\,v_x B \Rightarrow \begin{cases} \frac{dv_x}{dt} = \frac{qB}{\gamma m} v_y \\ \frac{dv_y}{dt} = -\frac{qB}{\gamma m} v_x \end{cases}$$

$$\frac{d}{dt}(v_x + i v_y) = \frac{qB}{\gamma m}(v_y - i v_x) = -i\omega(v_x + i v_y)$$

$$(v_x + i v_y)_t = e^{-i\omega t}(v_x + i v_y)_{t=0} = e^{-i\omega t} v_0 e^{i\alpha}$$

$$v_x(t) = \mathrm{Re}\left(v_x(t) + i v_y(t)\right) = v_x(0)\cos\omega t + v_y(0)\sin\omega t$$

$$v_y(t) = \mathrm{Im}\left(v_x(t) + i v_y(t)\right) = -v_x(0)\sin\omega t + v_y(0)\cos\omega t$$

rotational motion.

$$\boxed{\omega_B = \frac{qB}{\gamma m}}$$

Here $v_x(0) = v_0\cos\alpha$, $v_y(0) = v_0\sin\alpha$

$v_0 = \sqrt{v_x^2 + v_y^2}$

$$v_x(t) + i v_y(t) = v_0 \, e^{-i(\omega t - \alpha)}$$

$$v_x(t) = v_0 \cos(\omega t - \alpha) \quad , \quad v_y(t) = -v_0 \sin(\omega t - \alpha)$$

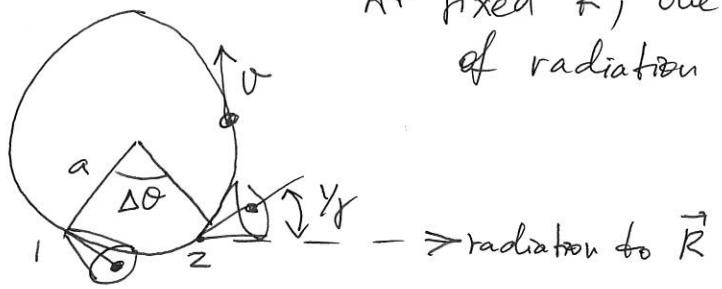
and v_0, α are determined by initial conditions.

$$v_x(t) = \frac{dx}{dt} = v_0 \cos(\omega t - \alpha) \Rightarrow x(t) = x_0 + \frac{v_0}{\omega} \sin(\omega t - \alpha)$$

$$y(t) = y_0 + \frac{v_0}{\omega} \cos(\omega t - \alpha)$$

Notice that $a = \frac{v_0}{\omega}$ is the radius of trajectory in x-y plane.

At fixed $\vec{R}$, one observes bursts of radiation emitted between points 1 and 2.



$$\Delta\theta = 2\cdot\frac{1}{\gamma} \Rightarrow \Delta s = a\cdot\Delta\theta = \frac{2a}{\gamma} = \frac{2v}{\gamma\omega_B}$$

But $v(t_2 - t_1) = \Delta s \Rightarrow t_2 - t_1 = \frac{2}{\gamma\omega_B}$

The arrival times $t_2^A - t_1^A = t_2 - t_1 - \frac{\Delta s}{c} = \frac{2}{\gamma\omega_B} - \frac{2v}{\gamma\omega_B c} =$

$$= \frac{2}{\gamma\omega_B}\left(1 - \frac{v}{c}\right) \approx \frac{1}{\omega_B}\frac{1}{\gamma^3}$$

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} \approx \frac{1}{\sqrt{2\left(1-\frac{v}{c}\right)}} \quad \text{for } \frac{v}{c} \sim 1$$

$$\Rightarrow 1 - \frac{v}{c} \approx \frac{1}{2\gamma^2} \quad \text{for } \gamma \gg 1.$$

$\Rightarrow$ temporal width of the observed pulses is smaller than $T \sim \frac{1}{\omega_B}$ by $\frac{1}{\gamma^3}$!

$\Rightarrow$ we expect broad spectral distribution up to frequencies $\omega_c \sim \omega_B \gamma^3 \gg \omega_B$!

Synchrotron Radiation

- Relativistically moving charge in magnetic field.
- Non-relativistic motion: cyclotron, $\omega_B = \frac{qB}{m}$ (essentially rotating dipole)
- Relativistic motion: ω_B and <u>its multiples</u>

because multipole expansion is based on small parameter

$\frac{a}{\lambda} \sim \frac{v}{c} \ll 1$. For relativistic motion $v \sim c$

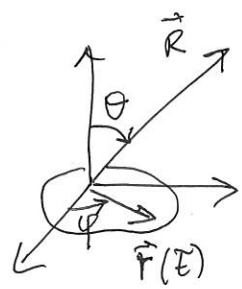
all terms in that expansion are of the same order $\Rightarrow$ multiple frequencies.

So need to start from the vector potential of relativistically moving charge

Lienard Wiechert potential

$$\vec{A}(\vec{R},t) = \frac{\mu_0 q}{4\pi R} \frac{\vec{v}(\tilde{t})}{1 - \frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}} \qquad \hat{n} = \frac{\vec{R}}{R}$$

valid for $R \gg \lambda \sim \frac{c}{\omega}$ (far field region)



<u>Important</u> $\tilde{t}$ = retarded time: light signal emitted by particle at $\vec{r}(\tilde{t})$ at time $\tilde{t}$ arrives at the detection point $\vec{R}$ at the present moment t:

$$c(t-\tilde{t}) = |\vec{R} - \vec{r}(\tilde{t})|.$$

Then $\vec{B}(\vec{R},t) = \vec{\nabla}_R \times \vec{A}(\vec{R},t)$.

$$\Rightarrow \text{obtain } \vec{B}(\vec{R},t) = \frac{\mu_0 q}{4\pi R} \vec{\nabla}_R \tilde{t} \times \frac{\partial}{\partial \tilde{t}} \frac{\vec{v}(\tilde{t})}{1 - \frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}}$$

$$c(t-\tilde{t}) = |\vec{R}-\vec{r}(\tilde{t})|$$

$$c(dt-d\tilde{t}) = d|\vec{R}-\vec{r}(\tilde{t})| =$$

But $d|\vec{R}-\tilde{r}|^2 = 2\,d(\vec{R}-\tilde{r})\cdot(\vec{R}-\tilde{r})$ $\Big\{$ vectors

Also $d|\vec{R}-\tilde{r}|^2 = 2\,|\vec{R}-\tilde{r}|\; d|\vec{R}-\tilde{r}|$ $\Big\{$ magnitudes

$$\Rightarrow \quad d|\vec{R}-\tilde{r}| = \frac{1}{|\vec{R}-\tilde{r}|}(\vec{R}-\tilde{r})\cdot\left(d\vec{R}-d\vec{r}(\tilde{t})\right)$$

But $\dfrac{\vec{R}-\tilde{r}}{|\vec{R}-\tilde{r}|} = \hat{n}$

$$\Rightarrow \quad d|\vec{R}-\vec{r}(\tilde{t})| = \hat{n}\cdot d\vec{R} - \hat{n}\cdot\underbrace{\vec{v}(\tilde{t})\,d\tilde{t}}_{d\vec{r}(\tilde{t})}$$

All spatial derivatives are taken at the same $t \Rightarrow dt=0$.

$$\Rightarrow \quad -c\,d\tilde{t} = \hat{n}\cdot d\vec{R} - \hat{n}\cdot\vec{v}(\tilde{t})\,d\tilde{t}$$

$$\Rightarrow \quad \left(c-\hat{n}\cdot\vec{v}(\tilde{t})\right)d\tilde{t} = -\hat{n}\cdot d\vec{R}$$

$$\Rightarrow \quad \vec{\nabla}_R\tilde{t} = \frac{d\tilde{t}}{d\vec{R}} = -\frac{\hat{n}}{c-\hat{n}\cdot\vec{v}(\tilde{t})} = -\frac{\hat{n}/c}{1-\hat{n}\cdot\vec{v}(\tilde{t})/c}$$

Thus $\vec{B}(\vec{R},t) = \vec{\nabla}_R\times\vec{A}(\vec{R},t) = \vec{\nabla}_R\times\left(\dfrac{\mu_0 q}{4\pi R}\,\dfrac{\vec{v}(\tilde{t})}{1-\frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}}\right)$

$$\approx \frac{\mu_0 q}{4\pi R}\,\vec{\nabla}_R\tilde{t}\times\frac{\partial}{\partial\tilde{t}}\,\frac{\vec{v}(\tilde{t})}{1-\frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}}$$

This is since $\vec{\nabla}_R\times\vec{f}(\tilde{t}) = \hat{e}_a\varepsilon_{abc}\dfrac{\partial}{\partial R_b}f_c(\tilde{t}) =$

$$= \hat{e}_a\,\varepsilon_{abc}\frac{\partial}{\partial\tilde{t}}f_c(\tilde{t})\,\frac{\partial\tilde{t}}{\partial R_b} = \hat{e}_a\,\varepsilon_{abc}\frac{\partial\tilde{t}}{\partial R_b}\,\frac{\partial f_c(\tilde{t})}{\partial\tilde{t}}$$

$$= \vec{\nabla}_R\tilde{t}\times\frac{\partial\vec{f}(\tilde{t})}{\partial\tilde{t}}$$

$$\vec{B}(\vec{R},t) = \frac{\mu_0 q}{4\pi R}\left(\frac{-\hat{n}/c}{1-\frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}}\right) \times \left\{ \frac{\dot{\vec{v}}}{1-\frac{\hat{n}\cdot\vec{v}}{c}} + \frac{\vec{v}\left(\frac{\hat{n}\cdot\dot{\vec{v}}}{c}\right)}{\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^2} \right\}$$

$$= \frac{\mu_0 q}{4\pi c R}\left\{ \frac{\dot{\vec{v}}\times\hat{n}}{\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^2} + \frac{(\vec{v}\times\hat{n})(\hat{n}\cdot\dot{\vec{v}})}{c\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^3} \right\}$$

$$= \frac{\mu_0 q}{4\pi c R}\left\{ \frac{\dot{\vec{v}}\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right) + \vec{v}\left(\frac{\hat{n}\cdot\dot{\vec{v}}}{c}\right)}{\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^3} \right\} \times \hat{n}$$

$$\dot{\vec{v}}(\hat{n}\cdot\vec{v}) - \vec{v}(\hat{n}\cdot\dot{\vec{v}}) = \hat{n}\times(\dot{\vec{v}}\times\vec{v}) = -v^2\,\hat{n}\times\vec{\omega}.$$

$$B\,A\,C \quad - C\,A\,B \quad = A\times(B\times C)$$

But $\underline{\dot{\vec{v}} = -\vec{\omega}\times\vec{v}} \Rightarrow \dot{\vec{v}}\times\vec{v} = (-\vec{\omega}\times\vec{v})\times\vec{v} = -v^2\vec{\omega}.$

$$+\vec{v}(\underbrace{\vec{\omega}\cdot\vec{v}}_{=\emptyset})$$

Thus $$\vec{B}(R,t) = \frac{\mu_0 q}{4\pi c R}\,\frac{\dot{\vec{v}} - \frac{v^2}{c}\,\hat{n}\times\vec{\omega}}{\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^3}\times\hat{n}$$

$$= \frac{\mu_0 q}{4\pi c R}(-)\,\frac{\vec{\omega}\times\vec{v} - \frac{v^2}{c}\,\vec{\omega}\times\hat{n}}{\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^3}\times\hat{n} = \qquad \left(\beta = \frac{v}{c} = \frac{a\omega}{c}\right)$$

$$= \frac{-\mu_0 q}{4\pi c R}\,\frac{\left[\vec{\omega}\times(\vec{v}-\beta v\hat{n})\right]\times\hat{n}}{\left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)^3}$$

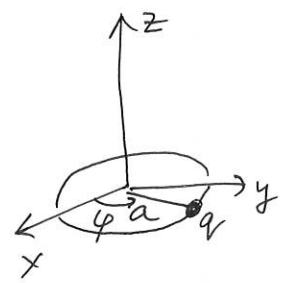
$$\frac{dP}{d\Omega} \sim \vec{B}^2 \sim \left([\vec{\omega} \times (\vec{v} - \beta v \hat{n})] \times \hat{n} \right)^2$$

Use $(\bar{A} \times \bar{B})^2 = A^2 B^2 - (\bar{A} \cdot \bar{B})^2$ then

$$\vec{B}^2 \sim \left[\vec{\omega} \times (\vec{v} - \beta v \hat{n})\right]^2 \underbrace{\hat{n}^2}_{=1} - \left((\vec{\omega} \times \vec{v} - \beta v \vec{\omega} \times \hat{n}) \cdot \hat{n} \right)^2 =$$

$$= \omega^2 (\vec{v} - \beta v \hat{n})^2 - \left(\vec{\omega} \cdot (\vec{v} - \beta v \hat{n})\right)^2 - \left(\vec{\omega} \cdot (\vec{v} \times \hat{n})\right)^2 .$$

Let $\vec{\omega} \parallel \hat{z}$ and $\hat{n} = (n_\perp, 0, n_z)$.



$$\varphi = \omega t + \varphi_0$$

$$\vec{v} \cdot \vec{n} = v n_\perp \cos\varphi \ , \ \vec{\omega} \cdot (\vec{v} \times \hat{n}) = \omega \hat{z} \cdot \left(v_x \hat{x} + v_y \hat{y} \right) \times \left(n_\perp \hat{x} + n_z \hat{z} \right)$$

$$= \omega \hat{z} \cdot \left(- v_x n_z \hat{y} - v_y n_\perp \hat{z} + v_y n_z \hat{x} \right) = - \omega n_\perp v \sin\varphi$$

Then we get $\omega^2 \left[v (\cos\varphi \hat{x} + \sin\varphi \hat{y}) - \beta v (n_\perp \hat{x} + n_z \hat{z}) \right]^2$

$$- \left(-\beta v \omega n_z \right)^2 - \left(\omega v n_\perp \sin\varphi \right)^2 =$$

$$= \omega^2 v^2 (\cos\varphi - \beta n_\perp)^2 + \underline{\omega^2 v^2 \sin^2\varphi} + \omega^2 \cancel{\beta^2 v^2} n_z^2$$

$$- \omega^2 \cancel{\beta^2 n_z^2} v^2 - \underline{\omega^2 v^2 n_\perp^2 \sin^2\varphi} =$$

$$= \omega^2 v^2 \left(\cos^2\varphi + \beta^2 n_\perp^2 - 2\beta n_\perp \cos\varphi \right) + \omega^2 v^2 \sin^2\varphi \underbrace{(1 - n_\perp^2)}_{n_z^2}$$

$$= \omega^2 v^2 \Big\{ (n_\perp^2 + n_z^2) \cos^2\varphi + n_z^2 \sin^2\varphi$$

$$+ \beta^2 n_\perp^2 - 2\beta n_\perp \cos\varphi \Big\} =$$

$$= \omega^2 v^2 \left(n_z^2 + \beta^2 n_\perp^2 + n_\perp^2 \cos^2\varphi - 2\beta n_\perp \cos\varphi \right)$$

Therefore $\frac{dP}{d\Omega} = \frac{c}{\mu_0} \vec{B}^2 R^2 = \frac{c}{\mu_0} \left(\frac{\mu_0 q}{4\pi c R}\right)^2 R^2 \cdot \omega^2 v^2 \cdot$

$$\cdot \frac{n_z^2 + \beta^2 n_\perp^2 - 2\beta n_\perp \cos\varphi + n_\perp^2 \cos^2\varphi}{(1-\beta n_\perp \cos\varphi)^6}$$

is the instantaneous power emitted into $d\Omega$.

We need to average it over the period of motion $\frac{2\pi}{\omega}$.

$$\overline{\frac{dP}{d\Omega}} = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} dt \, \frac{dP}{d\Omega} = \frac{\omega}{2\pi} \int_0^{2\pi} d\varphi \, \frac{dP}{d\Omega} \underbrace{\frac{dt}{d\varphi}}_{\frac{1}{\omega}\frac{dt}{d\tilde{t}}}$$

Recall $c(dt - d\tilde{t}) = d\left|\vec{R} - \vec{r}(\tilde{t})\right| =$

$$= \hat{n} \cdot d\vec{R} - \hat{n} \cdot \vec{v}(\tilde{t})\, d\tilde{t}$$

Now $d\vec{R} = 0 \Rightarrow dt = \left(1 - \frac{\hat{n}\cdot\vec{v}}{c}\right) d\tilde{t} = (1 - \beta n_\perp \cos\varphi)\, d\tilde{t}$

$$\overline{\frac{dP}{d\Omega}} = \frac{\mu_0 q^2 \omega^2 v^2}{16\pi^2 c} \int_0^{2\pi} \frac{d\varphi}{2\pi} \frac{n_z^2 + \beta^2 n_\perp^2 - 2\beta n_\perp \cos\varphi + n_\perp^2 \cos^2\varphi}{(1-\beta n_\perp \cos\varphi)^5}$$

This integration is done with the help of

$$\int_0^{2\pi} \frac{d\varphi}{2\pi} \frac{1}{a - b\cos\varphi} = \frac{1}{\sqrt{a^2 - b^2}} \quad \text{for } a = 1 > b = \beta n_\perp$$

$\beta \sim 1 \Rightarrow 1-\beta \sin\theta \cos\varphi$ is concentrated near $\theta \simeq \frac{\pi}{2}$, $\varphi \approx 0$

$$\sin\theta = \sin\left(\frac{\pi}{2}-\alpha\right) = \cos\alpha \approx 1-\frac{1}{2}\alpha^2$$

$$\frac{1}{1-\beta\left(1-\frac{1}{2}\alpha^2\right)\left(1-\frac{1}{2}\varphi^2\right)} = \frac{1}{1-\beta\left(1-\frac{1}{2}(\alpha^2+\varphi^2)\right)} = \frac{1}{1-\beta+\frac{\beta}{2}(\alpha^2+\varphi^2)}$$

$$\left(\gamma = \frac{1}{\sqrt{1-\beta^2}} \approx \frac{1}{\sqrt{2(1-\beta)}} \Rightarrow 1-\beta \approx \frac{1}{2\gamma^2}\right)$$

$$= \frac{2}{\gamma^2+\alpha^2+\varphi^2} \Rightarrow \text{angles } \boxed{\alpha, \varphi \sim \frac{1}{\gamma} \ll 1}$$

Observe that

$$\int_0^{2\pi} \frac{d\varphi}{2\pi} \frac{1}{(a-b\cos\varphi)^5} = \frac{1}{4!} \frac{\partial^4}{\partial a^4} \int_0^{2\pi} \frac{d\varphi}{2\pi} \frac{1}{a-b\cos\varphi} =$$

$$= \frac{1}{4!} \frac{\partial^4}{\partial a^4} \frac{1}{\sqrt{a^2-b^2}} = \frac{8a^4+24a^2b^2+3b^4}{8(a^2-b^2)^{9/2}}$$

$$\int_0^{2\pi} \frac{d\varphi}{2\pi} \frac{\cos\varphi}{(a-b\cos\varphi)^5} = -\frac{1}{4!} \frac{\partial^4}{\partial a^3 \partial b} \frac{1}{\sqrt{a^2-b^2}} =$$

$$= \frac{20a^3b+15ab^3}{8(a^2-b^2)^{9/2}}$$

$$\int_0^{2\pi} \frac{d\varphi}{2\pi} \frac{\cos^2\varphi}{(a-b\cos\varphi)^5} = \frac{1}{4!} \frac{\partial^4}{\partial a^2 \partial b^2} \frac{1}{\sqrt{a^2-b^2}} =$$

$$= \frac{4a^4+27a^2b^2+4b^4}{8(a^2-b^2)^{9/2}}$$

Adding everything up we get

$$\frac{\overline{dP}}{d\Omega} = \frac{\mu_0 c q^2 \omega^2 \beta^2}{32\pi^2} \frac{2-\sin^2\theta - \frac{1}{4}\beta^2(1+3\beta^2)\sin^4\theta}{(1-\beta^2\sin^2\theta)^{7/2}}$$

used $n_z = \cos\theta$

$n_\perp = \sin\theta$

Note that $\omega = \frac{qB}{m\gamma}$, $1/\gamma = \sqrt{1-\beta^2}$

Power radiated in all directions is

$$\overline{P} = 2\pi \underbrace{\int_0^{\pi} d\theta \sin\theta}_{-\int_1^{-1} d(\cos\theta) \ldots \,=\, \int_{-1}^{1} d(\cos\theta) \ldots} \frac{\overline{dP}}{d\Omega}$$

let $\cos\theta = x$, $\sin^2\theta = 1 - x^2$.

$$\overline{P} = \frac{\mu_0 c q^2 \omega^2 \beta^2}{32\pi^2} \cdot 2\pi \cdot 2 \underbrace{\int_0^1 dx \, \frac{2 - (1-x^2) - \frac{1}{4}\beta^2 (1+3\beta^2)(1-x^2)^2}{\left(1 - \beta^2 (1-x^2)\right)^{7/2}}}_{\text{(MATHEMATICA)} \quad \frac{4}{3(1-\beta^2)^2}}$$

$$\overline{P} = \frac{\mu_0 c q^2 \omega^2 \beta^2}{6\pi (1-\beta^2)^2}$$

This can be obtained differently, and without doing the integral

Section 4.8

K': $d\vec{P}' = 0$ since radiation is emitted into 4π.

emitted energy dW'.

K: $dW = \gamma\, dW'$, $dt = \gamma\, dt'$

$$\Rightarrow \frac{dW}{dt} = \frac{dW'}{dt'} \Rightarrow P = P'.$$

$\Rightarrow$ can use eqn (4.92) $P = \frac{2q^2}{3c^3}\left(\gamma^4 a_\perp^2 + \cancel{\gamma^6} a_\parallel\right)$

accelerations in frame K

$P \sim \frac{q^2 \gamma^4 a_\perp^2}{c}$ for synchrotron motion $(a_\parallel = 0)$

But $a_\perp = \frac{v^2}{a}$ $\Rightarrow P \sim q^2 \gamma^4 \frac{v^4}{c a^2} \sim \frac{c q^2 \left(\frac{v}{a}\right)^2 \left(\frac{v}{c}\right)^2}{(1-\beta^2)^2}$

radius

$$\Rightarrow P \sim \frac{c q^2 \omega^2 \beta^2}{(1-\beta^2)^2}$$

And in fact numerical coefficient $\left(\frac{\mu_0}{6\pi}\right)$ matches too.